SOPHIE OSTMEIER

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RESEARCH FOCUS

CS PhD Fellow at the ETH AI Center, working on making LLMs reliable enough for high-stakes settings such as medicine. Published in Nature, with first-author papers at ICML and EMNLP. Currently focusing on reliable agentic automation, i.e. agents that acquire new knowledge in unfamiliar situations and recover from adversarial data.

EDUCATION

ETH Zurich, Computer Science PhD Fellow at ETH AI Center — *Switzerland* 08/2026 - present
- Advised by Florian Tramèr and Yannis Chronis

Stanford University, Master's in Computer Science, AI specialization — *USA* 09/2024 - 09/2025
- GPA: 3.985/4

Technical University of Munich, MD — *Germany* 10/2015 - 11/2021
- Clinical Staatsexamen: Top 1.1% in Germany

SELECTED PUBLICATIONS (1,000+ citations, h-index 14)

Transformer architecture & LLM training

Blankemeier, L. . . . **Ostmeier, S.** et al. 2026. "Merlin: A Computed Tomography Vision-Language Foundation Model and Dataset". In: *Nature*

Ostmeier, S. et al. 2025. "LieRE: Lie Rotational Positional Encodings". In: *ICML*. GitHub

Delbrouck, J.-B. . . . **Ostmeier, S.** et al. 2025. "Automated Structured Radiology Report Generation". In: *ACL*

Ostmeier, S. et al. 2024. "GREEN: Generative Radiology Report Evaluation and Error Notation". In: *EMNLP, Findings*. GitHub

Detecting when models are wrong

Ostmeier, S. et al. 2026. "Gradient-Stable Attention Heads Signal LLM Correctness". In: *EMNLP, Findings*. GitHub

Varma, M. . . . **Ostmeier, S.** et al. 2026. "Symbal: Detecting Systematic Misalignments in Model-Generated Captions". In: *ICML*

Varma, M. . . . **Ostmeier, S.** et al. 2025. "TRoVe: Discovering Error-Inducing Static Feature Biases in Temporal Vision-Language Models". In: *NeurIPS*

Clinical deep learning

Ostmeier, S. et al. 2024. "Random expert sampling for deep learning segmentation of acute ischemic stroke on non-contrast CT". In: *Journal of NeuroInterventional Surgery*. US Patent

EXPERIENCE

University Hospital Zurich (USZ), Radiology Resident — *Switzerland* 10/2025 - 07/2026
- Completed neuroradiology rotation; part of emergency staff

Stanford University, Postgraduate Researcher, AIMI Center — *USA* 02/2022 - 09/2025
- Advised by Curtis Langlotz, Akshay Chaudhari and Jeremy Heit

IQVIA (formerly IMS Health), Working Student — *Germany* 01/2015 - 07/2017
- Healthcare consulting group, Munich

TECHNICAL SKILLS

ML: Python, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, SciPy, Pandas, R
Infrastructure: Git, Docker, Google Cloud Platform, SkyPilot, W&B, TensorBoard, SLURM

AWARDS & SCHOLARSHIPS

ETH AI Center PhD Fellowship — 08/2026 - present
DFG Walter Benjamin Fellowship, German Research Foundation — 05/2023 - 05/2025
DAAD RISE Worldwide Fellowship, German Academic Exchange Service — 07/2019 - 10/2019